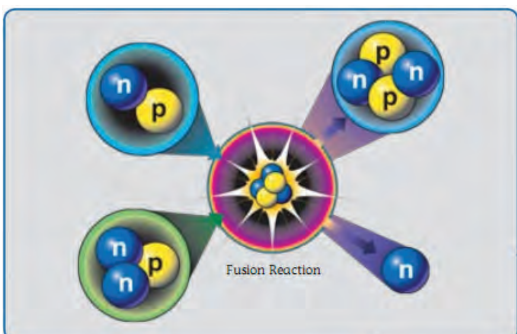
Water for sure
Segway inventor Dean Kamen has invented a filterless water purifier that makes drinking water from sewage, ocean water and urine, but it's reportedly very expensive

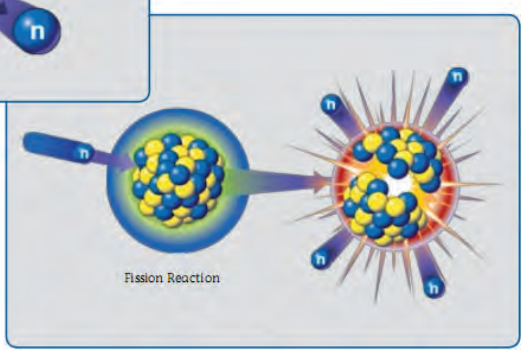
Patented bacteria
Coskata produces ethanol from landfill waste by processing waste gas in a bacteria-based bioreactor. The bacteria makes the method incredibly energy-efficient and cuts CO2 emissions by 84 per cent!

a horizontal axis in response to wind, generating electrical energy." says the official description. This electrical energy is transferred down the 1000-foot tether for immediate use, or to a set of batteries for later use, or to the power grid. Helium sustains MARS and allows it to ascend to a higher altitude than traditional wind turbines. MARS captures the energy available in the 600 to 1000-foot low level and nocturnal jet streams that exist almost everywhere.

Laser fusion possibilities

Harnessing the energy of the sun and stars to meet the Earth's energy needs has long been the holy grail of scientific and engineering endeavour. At the heart of this branch of alternative energy are ultra powerful lasers, since the kind of heat and pressure required to drive a fusion reaction can only be created by such lasers. A fusion power plant will generate energy by fusing atoms of deuterium and tritium – two isotopes of hydrogen, the lightest element. Deuterium can be extracted from abundant seawater, and tritium will be produced by the transmutation of lithium, a common element in soil. When the two atomic nuclei overcome their natural tendency to repel one another and fuse under the intense temperatures and pressures, a helium nucleus is formed and a small amount of mass lost in the reaction is converted to a large amount of energy according to Einstein's formula E=mc2. The National Ignition Facility based in California USA is at the forefront of this research, and released details on one of their press releases. "A fusion power plant would produce no greenhouse gas emissions, can operate continuously to meet demand, and would produce shorter-lived and easier-to-store radioactive by-products than current fission power plants" says the press release. Currently, estimates suggest that around 16 per cent of the world's electricity is generated by Nuclear Fission. Fission involves the splitting of heavy atoms such as uranium, to release energy. HiPER or High Power laser Energy Research facility is the European research project that runs along similar lines.





The difference between fusion and fission

Another Ambitious project called LIFE aims to create energy from Nuclear Waste. An acronym for Laser Inertial Fusion Engine, LIFE is an advanced energy concept under development at Lawrence Livermore National Laboratory, and is based on the physics and technology developed by NIF. LIFE has the potential to meet future worldwide energy needs in a safe, sustainable manner without carbon dioxide emissions. By burning nuclear waste for its fuel, LIFE may have the added benefit of reducing the planet's stockpile of spent nuclear fuel and the nuclear materials that lend themselves to nuclear proliferation. Once NIF's ultra powered laser completes a successful fusion burn (scheduled for 2010), it will serve as a springboard for LIFE, a hybrid technology that combines the best aspects of nuclear fusion, a clean, inherently safe and virtually unlimited energy source, with fission, a carbon-free, reliable energy technology, according to an NIF announcement.

digit
Watch out for more Green articles next month with a special surprise!!

Cold fusion to heat things up

Once relegated to the back alleys of Junk Science, Cold Fusion still remains that unattainable goal. Yet there is renewed interest in the technology and an imminent breakthrough may be on the horizon. Cold Fusion holds the promise of fusion at room temperature or near room temperature and that too on a table top. When it was announced back in 1989 more than twenty years ago by Martin Fleischmann and Stanley Pons, it shocked the world. But the whole thing was almost instantly discredited when other scientists tried to replicate the demonstration and the flash in the pan five minutes of fame it received came to an end. The Cold Fusion experiments as we know it are not strictly fusion. What transpires is more of a nuclear effect. Scientists bring together three components of this process – deuterium (which is abundant in sea water), palladium (a platinum like metal) and electricity. The resultant product is 'excess heat', which is what scientists are hoping to create – an output that is greater than the energy supplied to the experiment. The exact process of this nuclear effect is unknown, but stopping research just because something is not fully understood is not what scientists do. Researchers in this field found that there is indeed excess heat – about 20 times higher than the electricity input. However, this so called 'cold fusion' comes with its criticisms and sceptics. First of all, it does not have 100 per cent success rates, and critics are also skeptical about the way in which input/output energy is measured. However, reportedly, a US agency called Defense Advanced Research Projects Agency corroborated that there is indeed excess heat. There are lots of physicists and electro-chemists in labs all over the world researching this subject. As with any technology there is always scope for refinement, and when that eventuality happens, Cold Fusion could be the next big thing, that was revived from the ashes.